

Integration WS 10

Evaluate Q1-5

1. $\int \frac{x^2 dx}{\sqrt{25-x^2}}$

2. $\int \sqrt{x^2 + a^2} dx$

3. $\int \sqrt{a^2 - x^2} dx$

4. $\int \frac{x^2 dx}{\sqrt{x^2+4}}$

5. $\int \sqrt{4+x^2} dx$

6. a. If $n = \int_0^{\frac{\pi}{2}} x^n \cos x dx$, show that $I_n = \left(\frac{\pi}{2}\right)^n - n(n-1)I_{n-2}$

b. Hence find the value of I_4

c. Find the value of I_3 .

7. a. If $I_n = \int \frac{dx}{(x^2+1)^n}$ prove that $I_n =$

$$\frac{1}{2(n-1)} \left[\frac{x}{(x^2+1)^{n-1}} + (2n-3)I_{n-1} \right]$$

b. Hence evaluate $\int_0^1 \frac{dx}{(x^2+1)^2}$

8. If $I = \int_0^{\ln 2} \frac{e^x}{e^x + e^{-x}} dx$ and $J = \int_0^{\ln 2} \frac{e^{-x}}{e^x + e^{-x}} dx$,

find the exact values of $1+J$ and $1-J$ and

hence find the exact values of I and J .

9. a. If $I_n = \int_0^1 \frac{dx}{(1+x^2)^n}$, where n is a positive integer, show that $2nI_{n+1} = 2^{-n} + (2n-1)I_n$

b. Hence evaluate $\int_0^1 \frac{dx}{(1+x^2)^3}$

10. a. Prove that $\sqrt{\frac{1-x}{1+x}} = \frac{1-x}{\sqrt{1-x^2}}$

b. Hence, or otherwise evaluate $\int_0^{\frac{1}{2}} \sqrt{\frac{1-x}{1+x}} dx$

11. Let $I_n = \int_0^{\frac{\pi}{2}} \sin^n x dx$

a. Show that $I_n = \frac{n-1}{n} I_{n-2}$

b. Hence show that $\int_0^{\frac{\pi}{2}} \sin^{2n} x dx =$

$$\frac{\pi(2n)!}{2^{2n+1}(n!)^2}$$

12. Let $I_n = \int_0^1 (1-x^2)^n dx$, where n is a positive integer.

a. Show that $I_n = \frac{2n}{2n+1} I_{n-1}$

b. Find the value of I_3 .

13. a. Use the substitution $u = \frac{\pi}{2} - x$ to show that

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\tan x}{\tan x + \cot x} dx = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\cot x}{\tan x + \cot x} dx$$

b. Hence evaluate $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\tan x}{\tan x + \cot x} dx$

14. Let $I_n = \int_0^{\frac{\pi}{2}} \sin^n x dx$

a. Prove that $I_n = \frac{n-1}{n} I_{n-2}$, $n \geq 2$

b. Hence find I_5

c. Show that $I_{2m} = \frac{\pi(2m)!}{2^{2m+1}(m!)^2}$

15. Evaluate $\int_0^{\frac{\pi}{2}} \sqrt{\frac{1-\cos x}{1+\cos x}} dx$

16. Let $f(x) = \frac{x^2+9}{(x+2)(x^2+2x+5)}$

a. Find the values of the constants A, B and C so that $f(x) = \frac{A}{(x+2)} + \frac{Bx+C}{x^2+2x+5}$

b. Hence evaluate $\int_3^5 f(x) dx$

17. a. Use the substitution $x = \theta + \frac{\pi}{4}$ to show that

$$\int \frac{\cos x}{\cos x + \sin x} dx = \frac{1}{2} \int \frac{\cos \theta - \sin \theta}{\cos \theta} d\theta$$

b. Hence show that $\int_0^{\frac{\pi}{4}} \frac{1}{1+\tan x} dx = \frac{\pi}{16 \ln 2}$

18. a. Show that $\frac{d}{dx} \ln(x + \sqrt{x^2 + 4}) = \frac{1}{\sqrt{x^2 + 4}}$

b. Hence, or otherwise, show that

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{\sec^2 x}{\sqrt{\tan^2 x + 4}} dx = 2 \ln \frac{\sqrt{5}+1}{2}$$

19. Let $u_n = \int_0^{\frac{\pi}{4}} \tan^{2n} \theta d\theta$, where $n \geq 0$.

a. Show that $u_n = \frac{1}{2n-1} - u_{n-1}$, where $n \geq$

1.

b. Hence evaluate $\int_0^{\frac{\pi}{4}} \tan^6 \theta d\theta$.

20. a. Show that $\frac{1}{x^4+4} = \frac{x+2}{8(x^2+2x+2)} - \frac{x-2}{8(x^2-2x+2)}$

b. Consider the integral $I = \int_0^N \frac{1}{x^4+4} dx$, show

that $I = \frac{1}{16} \left[\ln \left(\frac{N^2+2N+2}{N^2-2N+2} \right) + 2 \tan^{-1}(N+1) + 2 \tan^{-1}(N-1) \right]$

c. Hence evaluate $\lim_{N \rightarrow \infty} \int_{-N}^N \frac{1}{x^4+4} dx$

21. Evaluate $\int_3^8 \frac{x}{(x+1)-\sqrt{x+1}} dx$ by using the substitution $x+1 = u^2$

22. Let $I_n = \int_0^{\frac{\pi}{2}} x^n \sin x dx$ for $n \geq 2$

a. Prove that $I_n = n \left(\frac{\pi}{2} \right)^{n-1} - n(n-1)I_{n-2}$

b. Hence evaluate $I_4 = \int_0^{\frac{\pi}{2}} x^4 \sin x dx$

23. Use the substitution $t = \tan \frac{\pi}{2}$ to evaluate

$$\int_0^{\frac{\pi}{3}} \frac{1}{2-\cos x} dx$$

24. a. Use a suitable substitution to help evaluate

$$\int_0^{\sqrt{2}} \frac{x^3}{\sqrt{x^2+1}} dx$$

b. How could this integral have been evaluated using just algebraic manipulation?

25. a. Use a trigonometric substitution to show that

$$\int_0^{\frac{1}{2}} \frac{x^2}{\sqrt{1-x^2}} dx = \frac{\pi}{12} - \frac{\sqrt{3}}{8}$$

b. How could this integral have been evaluated using just algebraic manipulation?

26. a. Use a suitable substitution to show that

$$\int_1^2 \sqrt{4-x^2} dx = \frac{2\pi}{3} - \frac{\sqrt{3}}{2}$$

b. Redo this problem by geometric means.

27. Consider the indefinite integral $I = \int \frac{dx}{x\sqrt{x^2-1}}$.

Clearly the domain of the integrand is disjoint, being $x > 1$ or $x < -1$. Thus it seems appropriate to use a different substitution in each part of the domain.

a. Find I for $x > 1$ by using the substitution

$$u = \sqrt{x^2-1}$$

b. Find I for $x < -1$ by using the

$$\text{substitution } u = -\sqrt{x^2-1}$$

28. a. Show that $(1+t^2)^{n-1} + t^2(1+t^2)^{n-1} = (1+t^2)^n$

b. Put $P_n = \int_0^x (1+t^2)^n dt$. Use integration by parts and the result in part (a) to show that

$$P_n = \frac{1}{2n+1} ((1+x^2)^n x + 2nP_{n-1}).$$

c. Hence determine P_4 : (i) by the reduction formula, (ii) by using the binomial theorem.

d. Hence write $1 + \frac{4}{3}x^2 + \frac{6}{5}x^4 + \frac{4}{7}x^6 + \frac{1}{9}x^8$ in powers of $(1+x^2)$

Answer

1. $\frac{1}{\sqrt{b-a^2}} \tan^{-1} \frac{x+a}{\sqrt{b-a^2}}$ if $b-a^2 > 0$, $-\frac{1}{x+a}$ if $b = a^2$ and

$$\frac{1}{2\sqrt{a^2-b}} \ln \left[\frac{x+a-\sqrt{a^2-b}}{x+a+\sqrt{a^2-b}} \right] \text{ if } b < a^2$$

2. $\tan^{-1}(3x-1)$

3. $2 \ln(x^2+2x+3) - \frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{x+1}{\sqrt{2}} \right)$

4. $\frac{25}{2} \sin^{-1} \frac{x}{5} - \frac{x}{2} \sqrt{25-x^2}$

5. $\frac{x}{2} \sqrt{a^2+x^2} + \frac{a^2}{2} \ln(x+\sqrt{x^2+a^2})$

6. b. $I_4 = \left(\frac{\pi}{2} \right)^4 - 12 \left(\frac{\pi}{2} \right)^2 + 12 \times 2I_0$;

c. $I_3 = \left(\frac{\pi}{2} \right)^3 - 6 \frac{\pi}{2} + 6$

7. b. $\frac{1}{6}$

8. $I = \ln \left(\sqrt{\frac{5}{2}} \right), J = \frac{2\sqrt{2}}{5}$

9. b. $\frac{1}{4} + \frac{3\pi}{32}$

10. b. $\frac{\pi}{6} - 1 + \frac{\sqrt{3}}{2}$

12. b. $\frac{16}{35}$

13. b. $\frac{\pi}{12}$

14. b. $\frac{8}{15}$; c. $\frac{\pi(2m)!}{2^{2m+1}(m!)^2}$

15. $\ln 2$

16. a. $f(x) = \frac{1}{x-2} - \frac{2}{x^2+2x+5}$;

b. $\ln 3 - \tan^{-1} \left(\frac{1}{7} \right)$

19. b. $\frac{13}{15} - \frac{\pi}{4}$

20. c. $\frac{\pi}{4}$

21. 7

22. b. $I_4 = \frac{\pi^3}{2} - 12\pi + 24$

23. $\frac{\pi\sqrt{3}}{6}$

24. a. $\frac{2}{3}$; b. use $x^3 = x(x^2+1) - x$

25. b. Begin by writing $x^2 = 1 - (1-x^2)$

26. b. The region is half a segment.

27. a. $\tan^{-1} \sqrt{x^2-1} + C_1$;

b. $\tan^{-1} \sqrt{x^2-1} + C_2$

28. d. $\frac{1}{9} ((1+x^2)^4 + \frac{8}{7}(1+x^2)^3 + \frac{48}{35}(1+x^2)^2 + \frac{192}{105}(1+x^2) + \frac{384}{105})$